

J-01-CX-geometric-depletion-kill-test v1: quantitative ω - S depletion without eigenframes

ZFL-CX

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1 Mission

$$\boxed{T^* < \infty \Rightarrow C_J}$$

$$\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_I^{freq} \wedge C_J \Rightarrow \int_0^{T^*} W_+ < \infty \text{ or } \perp}$$

Target:

$$W(t) = \int_{\mathbb{R}^3} |\omega|^2 \alpha \, dx, \quad \alpha = \xi^T S \xi, \quad \xi = \omega/|\omega|, \quad W_+ = \max(W, 0)$$

Goal quantitative:

$$\alpha_+ \lesssim \text{integrable quantity} \quad \Rightarrow \quad W_+ \leq \int |\omega|^2 \alpha_+$$

No eigenframes: we do not use $e_i(S)$ basis blocked in E-06/E-07. Only ξ vs S .

2 Definition C_J attempt

Under $T^* < \infty$, define depletion exponent:

$$C_J^{depl} : \exists \gamma < 1, \exists \rho(t) \downarrow 0, \quad |\alpha(x, t)| \leq C |\omega(x, t)|^\gamma \text{ on } \{|\omega| > \rho(t)^{-2}\}?$$

Or directional Lipschitz:

$$C_J^{Dir} : |\sin \varphi(\xi(x, t), \xi(y, t))| \leq \frac{|x - y|}{\rho(t)} \quad \text{with } \rho(t) \sim (T^* - t)^\beta$$

where φ angle between $\xi(x)$ and $\xi(y)$. This is Deng-Hou-Yu type, but we need it as *necessary* consequence of $T^* < \infty$, not hypothesis.

Proposed necessary candidate from $C_F^{conc} + C_I^{freq}$: Since $N_c \rightarrow \infty$ and $r_n \downarrow 0$, vorticity concentrates. Does concentration force alignment? Attempt: If $\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0$, then on high vorticity set $|\omega| \sim r_n^{-2}$, the direction ξ has bounded variation because $P_{N_n} u$ dominates.

Formally:

$$C_J : \exists t_n \uparrow T^*, \exists \Omega_n = \{|\omega(t_n)| \geq c r_n^{-2}\}, \quad \sup_{x, y \in \Omega_n} \|\xi(x, t_n) - \xi(y, t_n)\| \leq \delta_n \rightarrow 0?$$

3 Attempted $W_+ \leq f(C_J)$

We have identity $D_t|\omega| = \alpha|\omega|$. So $\alpha_+ = (\log|\omega|)_t$.

Thus

$$\int_0^{T^*} W_+(t) dt = \int_0^{T^*} \int |\omega|^2 \alpha_+ dx dt$$

If C_J^{Dir} holds with $\rho(t)$ integrable, then standard estimate (Constantin-Fefferman):

$$|S(x, t)| \lesssim \int \frac{|\omega(y)| |\sin \varphi|}{|x - y|^3} dy$$

Depletion: $|\sin \varphi| \leq |x - y|/\rho$ gives $|S| \lesssim \rho^{-1} \|\omega\|_{L^{3/2+}}$? Actually need quantitative:

$$\alpha_+(x, t) \leq C \int_{|x-y| \leq \rho} \frac{|\omega(y)|}{|x - y|^2} dy + C\rho^{-1} \dots$$

If $\rho(t) \geq c(T^* - t)^{1/2}$, then $\alpha_+ \in L_t^1 L^\infty$? Not proven.

Critical obstruction: C_J^{Dir} with $\rho(t) \rightarrow 0$ is compatible with $\int W_+ = +\infty$, because bound becomes worse as $\rho \rightarrow 0$.

To get $\int W_+ < \infty$, we would need $\rho(t) \geq \rho_0 > 0$ uniformly, which is false under blow-up, or $\|\omega(t)\|_{L^1}$ small, which contradicts C_F^{conc} .

Hence no uniform integrable bound.

4 Kill Test

J-01 KILLED if:

1. Recovers only Constantin-Fefferman / Deng-Hou-Yu qualitatively: ξ Lipschitz $\Rightarrow$ regular, but hypothesis not proved as necessary.
2. Requires geometric hypothesis not consequence of $T^* < \infty$: e.g., assumes $\|\nabla \xi\|_{L^\infty(\Omega_n)} \leq C$, or $\rho(t)$ lower bound, or Type I.
3. Controls α pointwise but not weighted by $|\omega|^2$: $|\alpha| \leq C$ does not imply $\int |\omega|^2 \alpha_+ < \infty$ because $|\omega|^2$ unbounded.
4. No temporally integrable quantity: $\alpha_+(t) \leq C(T^* - t)^{-1}$ still non-integrable for W_+ if $|\omega|^2 \sim (T^* - t)^{-2}$.
5. Compatible with $\int_0^{T^*} W_+ = +\infty$: C_J holds yet W_+ diverges, so no $\perp$.

Check: Our C_J candidate fails 1,2,5. Alignment from C_I^{freq} does not give uniform ρ_0 . The concentration $r_n \downarrow 0$ suggests $\rho(t) \sim r_n \rightarrow 0$, so estimate degenerates. No inequality $W_+ \leq f(C_J)$ with $\int f < \infty$.

Thus C_J^{depl} is not necessary; C_J^{Dir} is not necessary without extra assumption. Only qualitative statement: if ξ were uniformly Lipschitz then no blow-up, contrapositive gives ξ not uniformly Lipschitz under $T^* < \infty$, which is C_J^{neg} : loss of regularity of ξ , not depletion.

So $C_J^{neg} : \limsup_{t \uparrow T^*} \|\nabla \xi(t)\|_{L^\infty(\{|\omega| > M\})} = +\infty$ is necessary, but it does NOT bound W_+ above, it makes it worse.

5 Conclusion

$$C_J^{depl} \text{ not proven as necessary from } T^* < \infty$$

$$C_J^{neg} : \text{loss of direction regularity is necessary, but } \int W_+ = +\infty \text{ compatible}$$

$$\boxed{C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \wedge C_I^{freq} \wedge C_J^{neg} \not\approx \perp}$$

Classification: Genuinely new geometric family distinct from eigenframes, but KILLED as bridge to $\int W_+ < \infty$ or $\perp$. No quantitative $\alpha_+ \lesssim$ integrable.

$\mathcal{C}_{PROVEN}$ remains $\{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}\}$ plus C_J^{neg} as necessary loss, not depletion.

No J-02. Next family K should attempt quantitative CKN propagation of smallness in L^3 to bound W_+ via ε -regularity scale r_n .